Ex let $G = Z^2$ [under coordinate-wise +]
take $S = \{(1, 0), (0, 1)\}$

[what is $\ker(F_S \to Z^2)$?]

write $a = (1, 0)$ and $b = (0, 1)$
elts of F_S are words in a, b, a^{-1}, b^{-1}

if such a word contains
- M a's,
- N b's,
- M' a^{-1}'s,
- N' b^{-1}'s

then it is mapped to $(M - M', N - N')$ in Z^2

$\ker(F_S \to Z^2)$ = {words where the total exponents of a and b are both zero}

e.g., for any w, v in F_S, it contains
the commutator $[w, v] := wvw^{-1}v^{-1}$

[here w^{-1} means the inverse to w in F_S]

Fact ([follows from] Munkres 69.3–69.4)
- $\{[w, v] \mid w, v \text{ in } F_S\}$ is a generating set for $\ker(F_S \to Z^2)$
- the kernel is the smallest normal subgp containing $[a, b]$

altogether,

$$\mathbb{Z}^2 \simeq \langle a, b \mid aba^{-1}b^{-1} \rangle$$

via the identifications $a = (1, 0)$, $b = (0, 1)$

<u>Free Products</u> [goal: Seifert–van Kampen]

given $G_1 = \langle s \in S_1 \mid r \in R_1 \rangle$
$G_2 = \langle s \in S_2 \mid r \in R_2 \rangle$

<u>Df</u> the <u>free product</u> of G_1 and G_2 is

$$G_1 * G_2 = \langle s \in S_1 \sqcup S_2 \mid r \in R_1 \sqcup R_2 \rangle$$

[why well-defined?] $S_i \subseteq S_1 \cup S_2$
induces $F_{S_i} \subseteq F_{S_1 \cup S_2}$
so $R_i \subseteq F_{S_1 \cup S_2}$
so $R_1 \sqcup R_2 \subseteq F_{S_1 \cup S_2}$

turns out: $G_1 * G_2$ does not depend on how we present G_1 and G_2

<u>Rem</u> the free product $*$ is associative

<u>Ex</u> let $G = \{e, s\}$, the two-elt group
how to write down $G * G$? [pause]

need to distinguish two copies of s: say, “s” and “t”

$$G * G = \{e, s, t, st, ts, sts, tst, \ldots\}$$

(Munkres §70) Thm (Seifert–van Kampen)

suppose U_1, U_2 are open in X,

$X = U_1 \cup U_2$,

$A = U_1 \cap U_2$,

U_1, U_2, A are path connected

let $j_k : U_k \to X$ and $i_k : A \to U_k$ be the inclusion maps for $k = 1, 2$

then for any x in A:

1) the homomorphism
$\pi_1(U_1, x) * \pi_1(U_2, x) \to \pi_1(X, x)$
arising from $j_{1,*}$ and $j_{2,*}$ via the defn of free product is surjective

2) the kernel of the homomorphism is the smallest normal subgp of the domain containing the elts of the form

$$i_{1,*}([\gamma])^{-1}\, i_{2,*}([\beta])$$

as we run over elts $[\beta]$ in $\pi_1(A, x)$

[above, $i_{k,*}([\gamma])$ in $\pi_1(U_k, x)$, but then we implicitly embed it into the free product]

altogether [in the notation of group presentations]:

$$\pi_1(X, x) \simeq \langle [\gamma] \text{ in } \pi_1(U_1, x) * \pi_1(U_2, x) \mid i_{1,*}([\gamma])^{-1}\, i_{2,*}([\beta]) \text{ for } [\beta] \text{ in } \pi_1(A, x) \rangle$$